Refresh and Content Opportunity Scoring Using Historical Search Signals: A Leakage-Safe Grouped-Client Machine Learning Study

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Abstract Editorial teams responsible for large websites rarely have enough time to examine every page that loses or underuses search visibility. We therefore studied a narrower operational problem: which eligible pages should appear first in a human review queue when only historical, prediction-time search signals are available? The analysis covered 9,841,378 anonymized daily observations from March 2026 and 331,437 client-page pairs. After applying fixed coverage and visibility rules, 34,038 pages from 32 clients remained. Signals from March 1–15 were separated from a later-impression proxy defined over March 16–31. We compared a frozen position-adjusted CTR and visibility score with Logistic Regression and Random Forest on the same validation pages. The client-level holdout contained 24 training clients and eight unseen validation clients, with no overlap. At the first 50 ranks, Logistic Regression returned 32 proxy-positive pages (Precision@50 = 0.640), compared with 0.520 for the operational baseline and 0.540 for Random Forest. A deliberately invalid feature built from the outcome period raised Average Precision from 0.3939 to 0.9999, confirming that future information can make an evaluation appear almost perfect while rendering it unusable. The resulting artifact is a reason-coded review queue for editorial inspection. It is not an automatic refresh system, a causal estimate of editing impact, or evidence about a search engine's ranking algorithm.

Keywords search engine optimization, content opportunity scoring, content refresh prioritization, grouped validation, data leakage, logistic regression, Precision@K, ranking metrics, human-in-the-loop decision support

1 Introduction

Editorial review capacity is usually much smaller than the number of pages monitored by a large website. In that setting, a binary label for every page is less useful than an ordered shortlist that tells an analyst where to begin. Teams commonly sort by impressions, CTR, average position, or a hand-built opportunity score. These summaries are easy to explain, but each compresses a different part of page behavior and may miss combinations such as strong visibility, weak click capture, and unstable position.

The literature offers useful pieces of this problem. Organic-search studies have estimated CTR from observed clicks and impressions [1]; other work has compared technical and content attributes associated with ranking [2] or assessed webpages with supervised models [3]. Machine learning has also been applied to voice-search factors, query intent, and AI-supported SEO workflows [4–6]. These studies motivate the signals used here, but their objectives differ from a future-window review queue.

The evaluation design matters as much as the model. Pages owned by the same client can share templates, editorial routines, measurement settings, topic portfolios, and market conditions. A page-level random split may therefore place closely related observations on both sides of the experiment. Evidence from multi-source prediction shows why source-exclusive validation can provide a more defensible estimate for unseen groups [7], while analyses of conventional cross-validation warn against treating heterogeneous samples as interchangeable [8]. Leakage creates a second risk. A feature derived after the decision date, or a preprocessing step fitted before the split, can turn information from the answer into an input [9–11].

Against this background, we evaluated five practical questions. First, can historical search signals improve the first 50 positions of a review queue relative to a transparent baseline? Second, how should the result be interpreted when clients rather than pages define the holdout? Third, how much can an outcome-derived feature inflate the reported score? Fourth, which available signals does the retained model rely on? Finally, can the score be translated into review actions without allowing the model to edit content? The contribution is the combined validation and decision-support workflow. We do not propose a new learning algorithm.

2 Related Work and Research Gap

Prior research connects search visibility with clicks, impressions, technical attributes, content characteristics, and query context, but it does so through several different problem formulations. Strzelecki and Miklosik measured device-dependent organic CTR from actual search observations [1]. Cebeci and Diri combined technical audits, semantic attributes, regression, clustering, and importance analysis across 14,465 search-result items [2]. Ghattas *et al.* evaluated webpage performance with several supervised algorithms and a broad feature inventory [3]. Jallow and Seong, working on CTR prediction, emphasized interpretability alongside predictive performance [12]. Collectively, these studies justify using inspectable search and page signals, although none frames the output as a client-exclusive queue for later impression decline.

A second group of studies focuses on adjacent SEO decisions. Saeed *et al.* examined voice-query ranking factors using multiple classifiers [4]. Mohammadi *et al.* inferred query intent from search-result features [5]. Ziakis and Vlachopoulou surveyed AI uses in semantic search, content optimization, and technical auditing [6]; related ranking-factor research cautions that association should not be read as causation [13]. Nagpal and Petersen likewise showed that keyword choice depends on strategic trade-offs rather than a single relevance rule [14]. The common lesson is that SEO decisions are contextual. A score that is useful for prioritization should not be presented as a universal diagnosis.

Our validation choices draw on methodological literature outside SEO. Leakage has been documented as a source of optimistic and difficult-to-reproduce results in machine-learning science

Table 1: Dataset and Grouped Holdout

Item	Value
Daily source records	9,841,378
Unique client-page pairs	331,437
Eligible pages	34,038
Eligible clients	32
Training pages	18,075
Validation pages	15,963
Training clients	24
Validation clients	8
Client overlap	0
Validation positive rate	0.3008

[9], recommender evaluation [10], and transfer-learning pipelines [11]. Source-exclusive evaluation offers a closer match when the intended use concerns unseen entities [7]. User-held-out search reranking provides a related example in which the held-out entity reflects the deployment question [15]. Early-warning and triage systems also show why ranking metrics can be more useful than a single classification threshold when review capacity is limited [16, 17]. Human-in-the-loop reviews add an explicit boundary: the model organizes attention, while a person remains responsible for the action [18].

What remains underexplored is the combination of these elements in one reproducible SEO workflow. The reviewed literature provides limited evidence of a study that uses anonymized multi-client daily search data, fixes a decision date, constructs a later page-level proxy, separates clients across training and validation, compares against a frozen operational rule, demonstrates outcome leakage, evaluates the top of the queue, and then maps scores to review actions. This work addresses that combined methodological and operational gap. Its claims remain observational and specific to the reported validation design.

3 Materials and Methods

3.1 Data and Study Population

We used the March 2026 performance partition from the FlyRank ML Internship warehouse [19, 20]. It contains daily, anonymized client-page measurements rather than public webpages. Aggregation produced 331,437 distinct client-page pairs from 9,841,378 daily rows, with one modeling record per page within a client.

Eligibility rules were fixed before model fitting. A page needed at least 500 impressions in the feature window, five or more available measurement days in both windows, a defined CTR, and a mean position in the interval $(0, 20]$. We removed duplicate client-page records after aggregation. The final analytical population consisted of 34,038 pages across 32 clients. Names, domains, URLs, raw queries, credentials, and direct business identifiers were not used in the paper or the model.

3.2 Feature and Outcome Windows

We fixed the end of March 15 as the decision point. Every candidate predictor was computed from March 1–15, whereas the retrospective outcome was calculated from March 16–31. This ordering governed the entire pipeline: outcome-period values were excluded from imputation, scaling, expected-CTR references, feature construction, fitting, and model selection.

Table 2: Completed Prediction-Time Feature Set

Feature	Interpretation
$\log(1 + \text{impressions})$	Feature-window visibility volume
Clicks	Observed feature-window clicks
CTR	Clicks divided by impressions
Average position	Mean feature-window search position
Active days	Days with positive impressions
Position volatility	Standard deviation of daily position
Position band	Top 3, Page 1, or Page 2

For page i, the available-day-normalized impression rates were

$$\bar{I}_{i,F} = \frac{\sum_{t \in F} I_{it}}{D_{i,F}}, \tag{1}$$

$$\bar{I}_{i,O} = \frac{\sum_{t \in O} I_{it}}{D_{i,O}}, \tag{2}$$

where I_{it} is daily impressions and $D_{i,F}$ and $D_{i,O}$ are available measurement days in the feature and outcome windows.

3.3 Prediction-Time Features

Table 2 lists the seven predictors available at the decision point. We intentionally kept the feature set small enough to audit: six numerical summaries and one position-band category. Clicks, impressions, CTR, and position are established search-performance signals [1, 3]; active days and position variability describe coverage and instability within the same window. Median imputation, standardization, and categorical encoding were fitted only on training clients. Unseen position categories were ignored rather than inferred from the validation data.

An earlier proposal considered trend slopes, momentum, drawdown, repeated grouped validation, calibration, and client-block confidence intervals. We did not find completed, versioned outputs for those experiments in the repository used for this manuscript. They are therefore reported as planned extensions, not as evidence from the present study.

3.4 Later-Decline Proxy

The binary retrospective proxy was

$$y_i = \begin{cases} 1, & \bar{I}_{i,O} < 0.80\bar{I}_{i,F}, \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

The label records a relative change in measured visibility: a value of one means that the later daily impression rate fell below 80% of the earlier rate. It should not be read as a content-quality judgment. It also says nothing by itself about whether an edit is warranted or what effect an edit would have.

3.5 Frozen Operational Baseline

The baseline compared each page's CTR with a training-client median reference for its position band. Let c_i be observed CTR and $\tilde{c}_{b(i)}$ the training median for position band $b(i)$. Clipped CTR weakness was

$$g_i = \text{clip}\left(1 - \frac{c_i}{\tilde{c}_{b(i)}}, 0, 1\right). \tag{4}$$

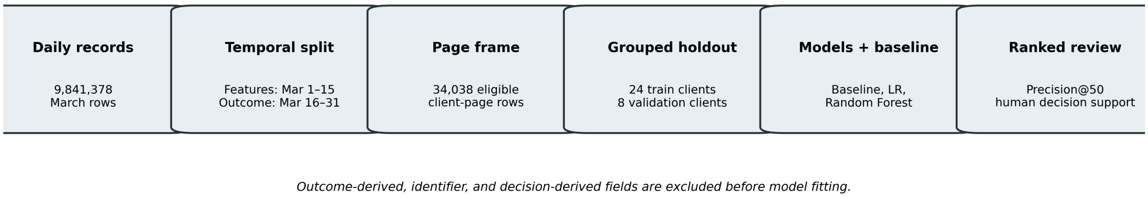

Figure 1: Leakage-safe grouped-client content opportunity scoring workflow. Outcome-period values are used only to construct the retrospective proxy and evaluate predictions.

Let q_i be the percentile of feature-window impressions relative to the training distribution. The frozen baseline score was

$$S_i^B = 100\,\mathbb{I}(g_i > 0)\,(0.70g_i + 0.30q_i)\,. \tag{5}$$

The position-band medians and visibility percentiles came only from the training clients. A page whose CTR met or exceeded its band reference was ineligible for a positive baseline score. We froze this rule before comparing it with the learned models.

3.6 Machine-Learning Models

We selected Logistic Regression as the primary learned model before examining the final comparison. Its output is naturally ordered as a probability score, and its compact form is easier to inspect than a larger nonlinear ensemble. For feature vector $\mathbf{x}_i$,

$$P(y_i = 1 \mid \mathbf{x}_i) = \frac{1}{1 + \exp[-(\beta_0 + \boldsymbol{\beta}^T \mathbf{x}_i)]}. \tag{6}$$

Class balancing was enabled, the iteration limit was 2,000, and all preprocessing remained inside the fitted pipeline.

Random Forest served as a controlled check for nonlinear interactions. We used 300 trees, square-root feature sampling, balanced subsampling, and a minimum leaf size of 10. Tree ensembles have been used in related ranking and webpage-performance studies [2, 3], but we did not assume that extra flexibility was automatically useful. The ensemble would replace Logistic Regression only if Precision@50 improved by at least 0.05 and AP did not decline. The measured results did not meet that rule.

3.7 Grouped-Client Validation

Clients, rather than individual pages, defined the retained holdout. Using `GroupShuffleSplit` with a 25% validation fraction and seed 42 placed every page from a client on one side of the split. This follows the leave-source-out principle used when the intended generalization target is a new source [7] and avoids assuming that heterogeneous groups are exchangeable [8]. The resulting partition contained 24 training clients and eight validation clients; their intersection was empty.

We also ran a stratified random-row split with the same seed and fraction as a contrast. Because clients appeared in both partitions, that result answers a different and weaker generalization question. It was retained for audit purposes, not as the final estimate. Figure 1 shows the sequence of decisions.

3.8 Leakage Audit

Before fitting, we reviewed each field against the question, "Could this value have been known at the end of March 15?" Outcome impressions, outcome coverage, the label, downstream product scores, action flags, identifiers, and privacy-sensitive fields failed that test and were excluded. This availability-based check follows leakage frameworks that examine feature construction, preprocessing, splitting, and evaluation rather than only the final training matrix [9–11]. To make the risk visible, we then created one deliberately invalid ratio:

$$r_i^{\text{leak}} = \frac{\bar{I}_{i,O}}{\bar{I}_{i,F}}. \tag{7}$$

Equation (7) contains the period used to define the label in Eq. (3). A model that scores nearly perfectly with this input has learned access to the answer; the result is therefore a failed validity check, not an improved predictor.

3.9 Evaluation Metrics

Precision@50 was chosen before the final comparison because the intended user reviews a small queue rather than every page. Similar capacity constraints motivate rank-focused evaluation in early-warning and triage applications [16, 17]. For the set $\mathcal{T}_K$ of the K highest-scored validation pages,

$$P@K = \frac{1}{K} \sum_{i \in \mathcal{T}_K} y_i. \tag{8}$$

Recall@K was

$$R@K = \frac{\sum_{i \in \mathcal{T}_K} y_i}{\sum_{i=1}^{n} y_i}. \tag{9}$$

The repository's lift field was an absolute difference, not a multiplicative ratio:

$$\Delta@K = P@K - \pi, \tag{10}$$

where π denotes the positive rate in the validation set. AP summarizes the ordering of positive pages over the full ranked list, while ROC-AUC measures pairwise discrimination. We also calculated threshold-based accuracy, class precision, recall, and

$$F_1 = \frac{2PR}{P + R} \tag{11}$$

at a threshold of 0.50. These threshold results are diagnostic; they do not replace the queue-oriented primary measure.

4 Experimental Setup

The analysis ran in a Python 3.12 environment. Exact package versions, the random seed, model settings, executed notebooks, fitted pipeline, and aggregate receipts are recorded in the public repository [20]. DuckDB was used for warehouse aggregation, pandas and NumPy for tabular processing, scikit-learn for the models and evaluation, Matplotlib for figures, and joblib for the serialized pipeline. The read credential required for the gated warehouse was stored outside the notebook and is not part of the repository.

Algorithm 1 records the executed sequence. Importantly, the baseline and both learned models were scored on the same grouped validation rows.

Algorithm 1 Leakage-Safe Grouped-Client Content Opportunity Scoring

Require: Daily anonymized client-page performance records
Ensure: Ranked human-review queue

```
Select the March 2026 partition
Separate feature and outcome windows
Aggregate daily records to client-page rows
Apply fixed eligibility criteria
Construct prediction-time features
Construct the later-decline proxy
Remove future, target-derived, decision-derived, identifier, and privacy-sensitive fields
Split pages by client into training and validation
Fit imputation, scaling, and encoding on training clients
Build the frozen CTR-weakness and visibility baseline
Train Logistic Regression and Random Forest
Score identical validation rows
Compute Precision@K and secondary metrics
Execute and reject the deliberate leakage experiment
Inspect permutation importance and failure examples
Apply the predeclared model-complexity rule
Assign human-readable review actions and limitations
Export the ranked queue for human review
```

Table 3: Grouped-Client Model Comparison on Identical Validation Rows

Method	Base rate	P@20	P@50	Absolute lift@50	R@50	AP	ROC-AUC
ML-07 baseline	0.3008	0.550	0.520	0.2192	0.0054	0.3776	0.6059
Logistic Regression	0.3008	0.500	**0.640**	**0.3392**	0.0067	**0.3939**	0.6175
Random Forest	0.3008	0.500	0.540	0.2392	0.0056	0.3908	**0.6206**

5 Results and Discussion

5.1 Model-versus-Baseline Performance

The three methods in Table 3 and Fig. 2 were evaluated on exactly the same 15,963 validation pages. Logistic Regression placed 32 proxy-positive pages among the first 50 ranks, giving Precision@50 = 0.640. The frozen baseline returned 0.520 and Random Forest 0.540. Relative to the baseline, the selected model added six proxy-positive pages to a 50-page review batch, an absolute Precision@50 difference of 0.120. Its absolute difference from validation prevalence was 0.3392.

Random Forest recorded the largest ROC-AUC, although the margin over Logistic Regression was small. That advantage did not carry into the top 50 ranks, and AP was also lower. Under the predeclared rule, the ensemble therefore remained a comparison model. This is a useful reminder that the chosen metric should reflect the work the system is expected to support.

5.2 Random-Row versus Grouped Validation

The random-row contrast in Table 4 included 26 validation clients, all of whom were also represented in training. It produced Precision@50 = 0.520. The client-exclusive holdout produced 0.640 with no overlap. In this run the grouped estimate was higher, so the data do not support the routine claim that random splitting must inflate every metric. The important distinction is conceptual: the random split tests new pages from partly familiar clients, whereas the grouped split tests pages belonging to clients not seen during fitting.

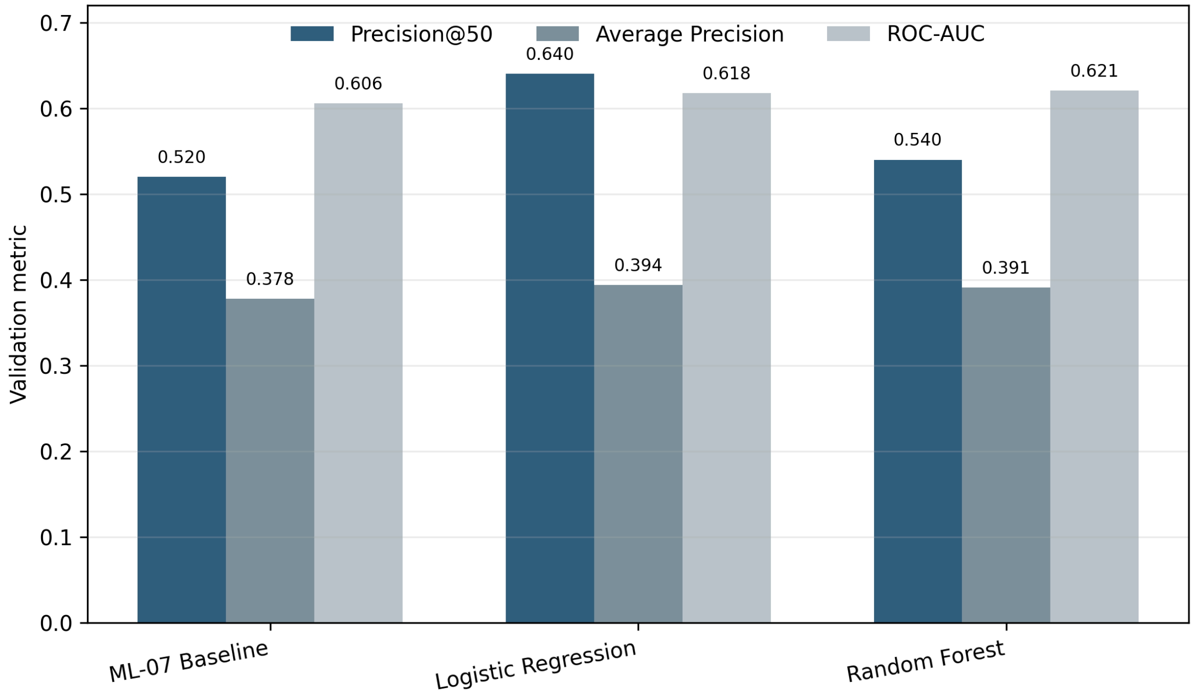

Figure 2: Grouped-client model and baseline comparison. Precision@50 is the primary operational metric; AP and ROC-AUC are secondary.

Table 4: Validation-Design Contrast

Design	Rows	Clients	Overlap	P@50	AP
Random row	8,510	26	26	0.520	0.4011
Grouped client	15,963	8	0	0.640	0.3939

5.3 Leakage Experiment

Adding the outcome-derived ratio changed the evaluation dramatically: Precision@50 and ROC-AUC reached 1.000, and AP reached 0.9999. The honest model produced 0.640, 0.6175, and 0.3939, respectively. Figure 4 visualizes the difference. No substantive conclusion can be drawn from the larger scores because the added feature includes the future period that determines the proxy. Here, near-perfect performance is diagnostic evidence that the experiment is invalid.

5.4 Feature Interpretation

Permutation tests on the grouped holdout showed the largest AP decreases when CTR, position band, average position, and position volatility were disrupted. These are measurable signals that a reviewer can inspect, which is consistent with the interpretability emphasis in recent ranking and CTR studies [2, 12]. Table 6 and Fig. 5 give the mean decrease and its repeated-permutation variability.

This result describes dependence of one fitted model on one held-out partition. It is not an intervention result. Correlated predictors may divide or exchange importance, and changing CTR or position is not guaranteed to change the later proxy.

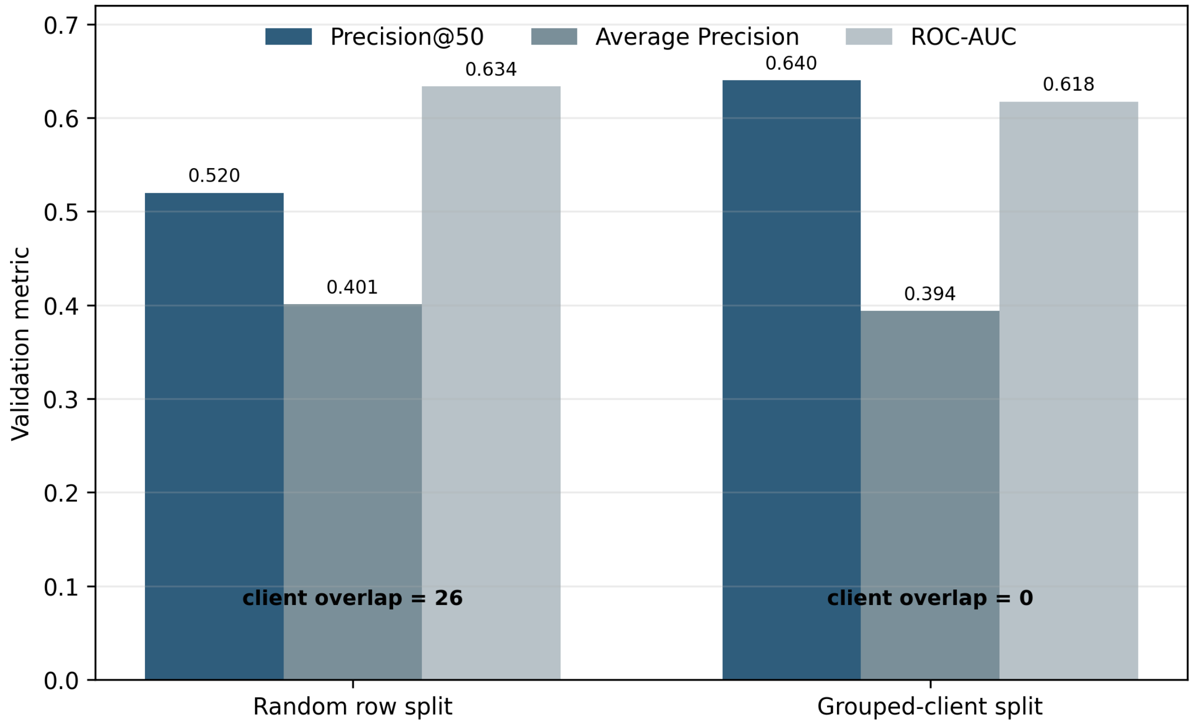

Figure 3: Measured random-row and grouped-client results. The grouped design is retained because it evaluates unseen clients.

Table 5: Honest and Deliberately Leaky Feature Sets

Feature set	P@50	AP	ROC-AUC
Honest prediction-time	0.640	0.3939	0.6175
Leaky + decline ratio	1.000	0.9999	1.0000

5.5 Threshold and Error Analysis

Using 0.50 only as a diagnostic threshold, the model produced 5,880 true negatives, 5,282 false positives, 1,705 false negatives, and 3,096 true positives. Overall accuracy was 0.562, whereas predicting the majority class for every row would yield 0.699. Macro F_1 was 0.549 and weighted F_1 was 0.580; the positive and negative class F_1 values were 0.470 and 0.627. These figures show why the model should not be described as a general binary classifier.

Within the operational queue, 18 of the first 50 pages did not meet the later-decline proxy. Several of these false positives had no observed clicks despite Page-1 positions and also showed substantial position variability. False negatives near the 0.50 threshold included high-volume pages whose positions looked comparatively stable during the feature window. Neither pattern can be resolved from the model inputs alone. A reviewer would need to check tracking quality, query mix, seasonality, result-page changes, and competitor movement before deciding whether the page deserves an edit.

6 Human-Reviewed Content Action Playbook

A probability by itself is not an editorial instruction. We therefore converted the retained score into a nonproduction playbook whose purpose is to organize review. This follows the human-in-the-loop pattern used in other prioritization systems: the model can order cases, but a domain

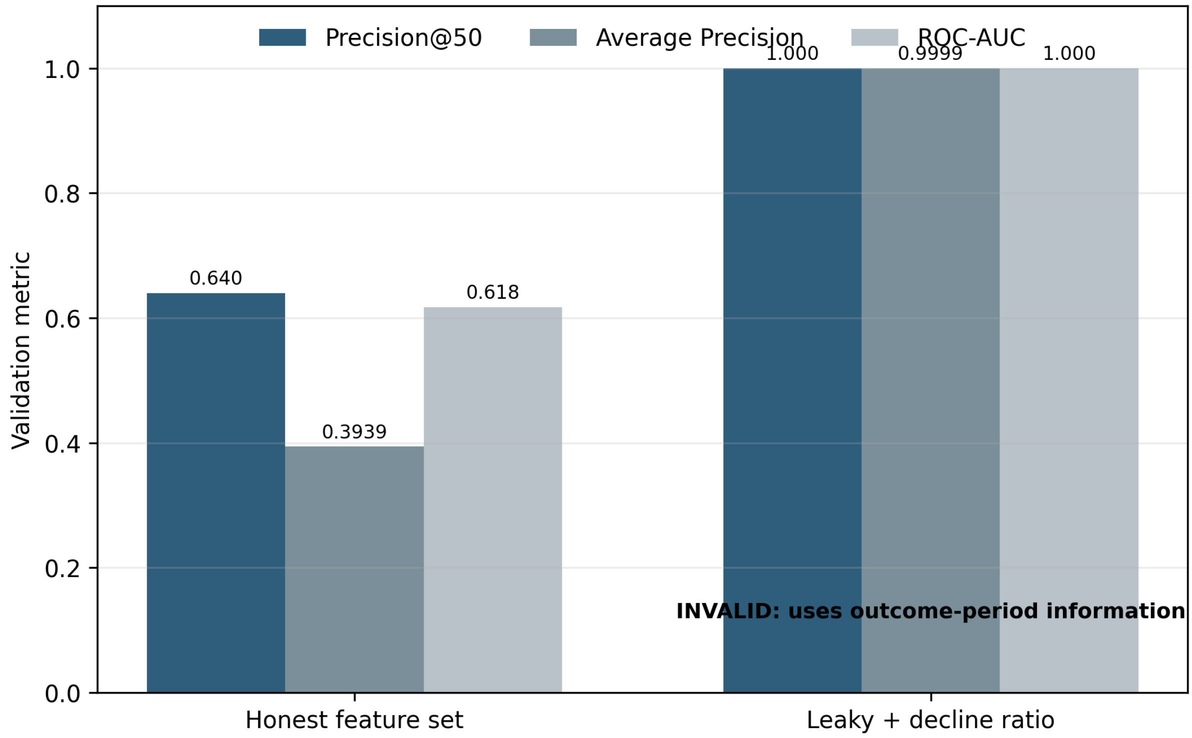

Figure 4: The deliberate future-derived feature produces invalid near-perfect metrics, motivating strict temporal feature availability checks.

reviewer owns the decision [17, 18]. Available actions included CTR review, expansion or optimization review, refresh review, refresh-and-protect review, monitoring, protection, manual review, and no immediate action. Reason codes recorded the measured conditions behind an action so that a reviewer could disagree with it.

Approval required checks beyond the seven model features. The reviewer had to examine data availability, indexing and canonical status, query intent, search-result features, seasonality, competing pages, factual accuracy, business importance, legal restrictions, the measurement window, and a rollback condition. These checks cover information that was absent from the modeling frame. The workflow never rewrites, publishes, deletes, merges, redirects, canonicalizes, prunes, or no-indexes a page. Figure 7 marks this boundary.

7 Limitations and Threats to Validity

Internal validity: The reported comparison rests on one grouped holdout and one 80% proxy threshold. We controlled the leakage pathways we could identify, but the model cannot observe demand shocks, indexing events, competitor changes, or changes to the result page. The deliberate leaked feature demonstrates one failure mode; it is not a complete catalogue of contamination risks.

Construct validity: The target is a change in impressions, not a direct measure of content decay, factual quality, or editorial need. Average position blends queries and devices, and CTR can move when the query mix or result-page layout changes. A positive proxy should therefore be interpreted as a reason to inspect a page, not as a diagnosis.

External validity: The development period covers a single month and 32 eligible clients. The result may differ by language, industry, season, market, or search environment. Because only one grouped partition was retained, the study does not estimate how sensitive Precision@50 is

Table 6: Held-Out Permutation Importance

Feature	Mean	Std. dev.
Feature CTR	0.07179	0.00368
Position band	0.03623	0.00237
Average position	0.01659	0.00170
Position volatility	0.01167	0.00181
Feature clicks	0.00022	0.00013
Log impressions	0.00020	0.00088
Active days	−0.00151	0.00044

Table 7: Human-Reviewed Action Archetypes

Action	Required human interpretation
CTR review	Inspect title, description, intent, SERP features, and tracking
Expand / optimize	Verify content gap, demand, and business relevance
Refresh review	Check factual age, decay evidence, and existing strengths
Protect / monitor	Avoid damaging a stable or high-value page
Manual review	Resolve inadequate coverage or scope uncertainty
No immediate action	Reassess after a defined monitoring interval

to a different set of held-out clients.

Operational validity: The warehouse extract did not provide the model with raw query text, competitor observations, conversions, revenue, direct content-quality labels, or intervention outcomes. The threshold accuracy was lower than majority accuracy. That result limits the use case to ranked review and argues against automatic classification or editing.

The repository used for this paper also lacks completed experiments for temporal slopes, repeated grouped cross-validation, rolling time evaluation, probability calibration, client-block confidence intervals, and top-K stability. We have not reported those planned analyses as completed findings.

8 Conclusion and Future Work

On the retained client-exclusive holdout, Logistic Regression improved the first 50 positions of the review queue relative to the frozen CTR-weakness and visibility baseline. Precision@50 increased from 0.520 to 0.640, with no client shared between training and validation. Random Forest offered a slightly larger ROC-AUC but placed fewer proxy-positive pages in the first 50 ranks and did not satisfy the predeclared complexity rule.

The leakage audit was equally important to the substantive result. A ratio computed from the outcome period raised AP from 0.3939 to 0.9999 because it nearly encoded the label. Rejecting that result illustrates why feature availability must be checked against the decision date rather than only against a column-name list.

The practical contribution is a public-safe queue for human review. Within the March 2026 design, the score can help allocate attention; it cannot show that a page must be refreshed, that an edit will recover traffic, or that the model has identified a search engine rule.

A stronger follow-up study should evaluate impression, click, CTR, and position trends; early-versus-late changes; active-day and zero-click ratios; repeated client-level splits; rolling temporal tests; calibration; alternative proxy thresholds; and confidence intervals that resample clients rather than pages. Privacy-safe metadata, multilingual evaluation, and controlled refresh

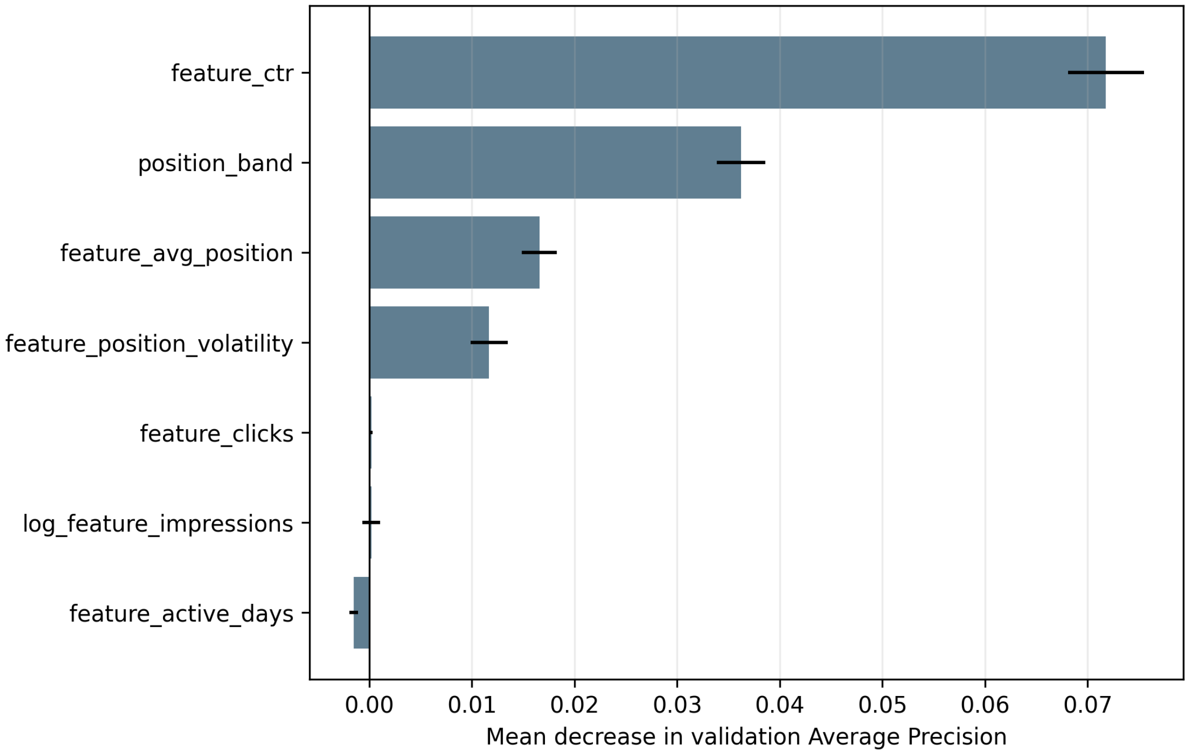

Figure 5: Grouped-holdout permutation importance measured as the decrease in Average Precision. Importance reflects model reliance, not causal effect.

interventions would address different aspects of generalization and causality.

Supplementary information

No supplementary information is included with this manuscript. The public repository contains aggregate reproducibility receipts, code, and publication-safe figures.

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Generative AI statement

Generative AI tools were used during early code drafting and for limited suggestions on manuscript organization and language. The authors subsequently rewrote the narrative, checked the numerical claims against executed notebooks and receipts, reviewed the cited sources, and accept responsibility for the submitted text and analysis.

Declarations

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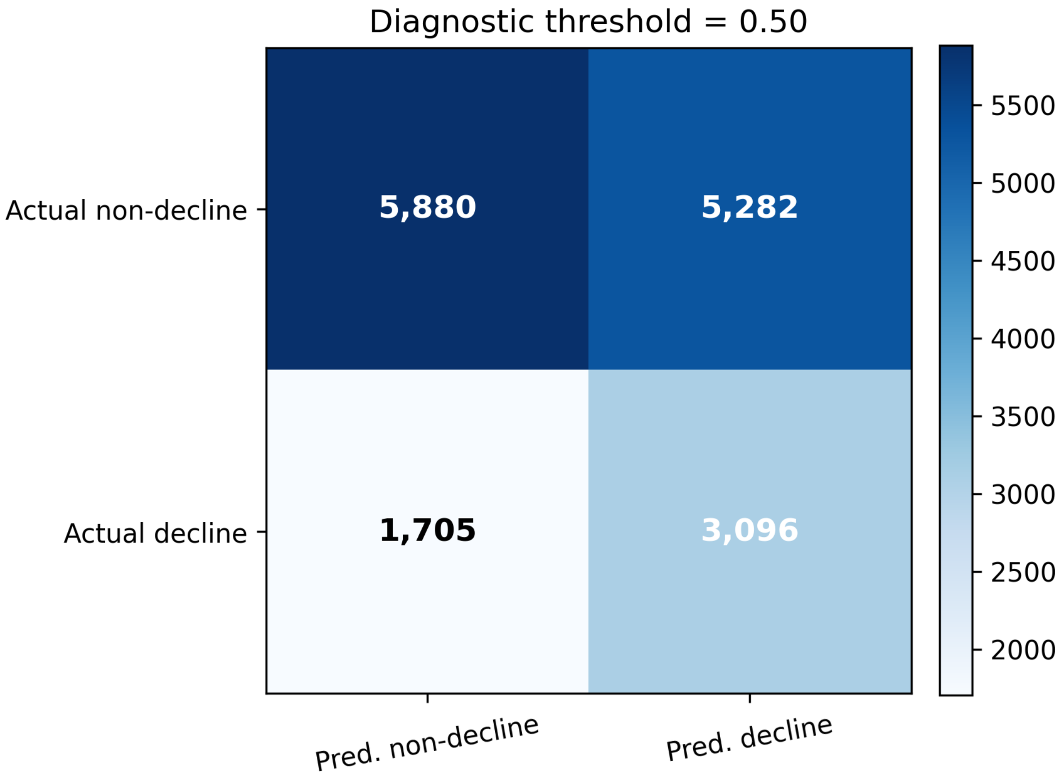

Figure 6: Diagnostic confusion matrix at threshold 0.50. Threshold classification is secondary to the ranked review objective.

Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

Not applicable. The study used anonymized, non-human search-performance records and did not involve human participants, human samples, or animals.

Consent for publication

Not applicable.

Data availability

The raw FlyRank warehouse is gated and is not redistributed because it contains protected operational data. Public-safe aggregate results, executed notebooks, figures, and reproducibility receipts are available in the project repository at `https://github.com/Zafar488/flyrank-ml-internship`. Access to the underlying dataset is governed by FlyRank's data-access conditions.

Materials availability

Not applicable.

Code availability

The analysis notebooks, aggregate receipts, and publication-safe code are available at `https://github.com/Zafar488/flyrank-ml-internship`.

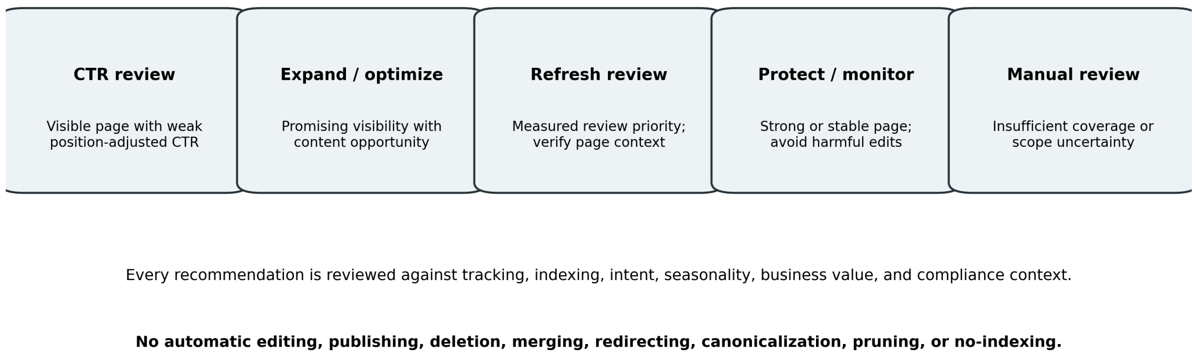

Figure 7: Action categories and the mandatory review boundary. Model output prioritizes inspection but does not authorize a page change.

Author contributions

Z.U. conceived the study, implemented the analysis, conducted validation and interpretation, prepared the visualizations, and drafted the manuscript. A.A.K. contributed to literature review, methodological discussion, and manuscript review. Both authors reviewed and approved the final manuscript.

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